Question # 5 of 5 (Start time: 05:19:48 PM, 07 December 2022) **Total Marks: 1**

Which of the following is the example of a contravariant vector.

Select the correct option

Reload Math Equations

- Velocity vector to a curve
- Gradient vector to a surface

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz #1) Quiz Start Time: 05:18 PM, 07 December 2022

Question # 4 of 5 (Start time: 05:19:39 PM, 07 December 2022) Total Marks: 1

A vector whose _____transform in a contravariant fashion is a contravariant vector.

Select the correct option

Reload Math Equations

- direction
- components

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz #1) Quiz Start Time: 05:18 PM, 07 December 2022

Question # 3 of 5 (start time: 05:19:15 PM, 07 December 2022) Total Marks: 1

In Cartesian coordinate system the covariant and contravariant components of a vector are different.

Select the correct option

Reload Math Equations

- true
- false

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and Its Applications (Quiz #1) Quiz Start Time: 05:18 PM, 07 December 2022

Question # 2 of 5 (start time: 05:18:52 PM, 07 December 2022) Total Marks: 1

Covectors are also called_____.

Select the correct option

Reload Math Equations

- dual vectors
- unit vectors

Click to Save Answer & Move to Next Question

PIU EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1) Quiz Start Time: 05:18 PM, 07 December 2022

Question # 1 of 5 (Start time: 05:18:42 PM, 07 December 2022) Total Marks: 1

Covariant components of a vector represent its _____

Select the correct option

Reload Math Equations

- perpendicular projections
- parallel projections

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

A contravariant vector is often abbreviated as_____.

Select the correct option

- vector
- velocity vector



Question # 1 of 5 (Start time: 03:34:40 PM, 07 December 2022)

Which one of the following is the contravariant tensor of order 2.

Select the correct option

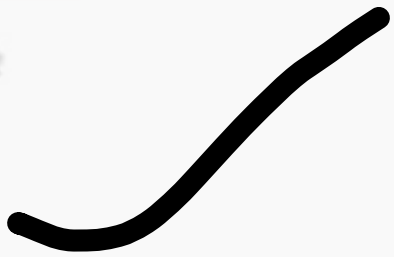
○	T^{ij}
○	A^{ii}
○	A^{i}_{i}
○	T_{ij}

EASY STUDY EASY SOLUTION

The order of _____ is 3.

Select the correct option

- B^{lm}_{mnk}



not sure

- A^{ij}_{j}

EASY STUDY EASY SOLUTION

Question # 3 of 5 (Start time: 04:28:12 PM, 07 December 2022)

The transformation law for a covariant vector is given by

Select the correct option

○	$\bar{A}_i = A_i \frac{\partial x^j}{\partial \bar{x}^j}$
○	$\bar{A}_i = A_j \frac{\partial x^j}{\partial \bar{x}^i}$



PUC EASY STUDY EASY SOLUTION

Question # 5 of 5 (Start time: 04:29:36 PM, 07 December 2022)

If $\bar{x}^1 = 2x^1$ and $\bar{x}^2 = x^2$ then $x^1 =$_____ in terms of tilde coordinates.

Select the correct option

- $\frac{1}{2}\bar{x}^1$
- $2\bar{x}^1$

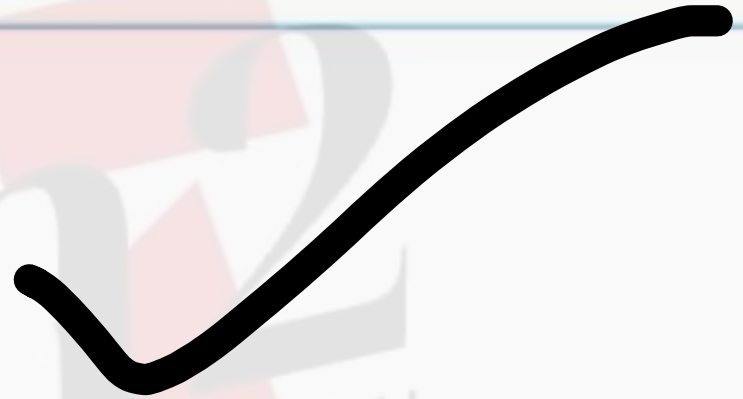
EASY STUDY EASY SOLUTION

Question # 1 of 5 (Start time: 04:30:52 PM, 07 December 2022)

Using the transformation law $\bar{x}_1 = 2x_1$ and $\bar{x}_2 = x_2$ find $\frac{\partial x^1}{\partial \bar{x}^1}$

Select the correct option

- ○ 1/2
- ○ 1



PIU EASY STUDY EASY SOLUTION

uestion # 4 of 5 (Start time: 04:34:51 PM, 07 December 2022)

_____ of a vector represent its parallel projections onto coordinate axes.

lect the correct option

- Contravariant components
- Covariant components

PUC EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 1 of 5 (Start time: 04:39:25 PM, 07 December 2022)

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x^1}{\partial \tilde{x}^1}$?

not sure

Select the correct option

- $-\cos\theta$
- $\cos\theta$



PUC EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 4 of 5 (Start time: 04:42:53 PM, 07 December 2022)

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x^1}{\partial \tilde{x}^2}$

Select the correct option

- ○ $r \sin \theta$
- ○ $-r \sin \theta$



PIE EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #

Question # 5 of 5 (Start time: 04:43:43 PM, 07 Decemb

Covectors are also called_____.

Select the correct option

○	dual vectors
○	unit vectors



EASY STUDY EASY SOLUTION

Question # 3 of 5 (Start time: 04:47:03 PM, 07 December 2022)

Which one of the following is the covaraint tensor of order 3?

Select the correct option

- T_{ijk}
- A^{m}_{kl}
- T^{ijk}
- A^{j}_{ij}

EASY STUDY

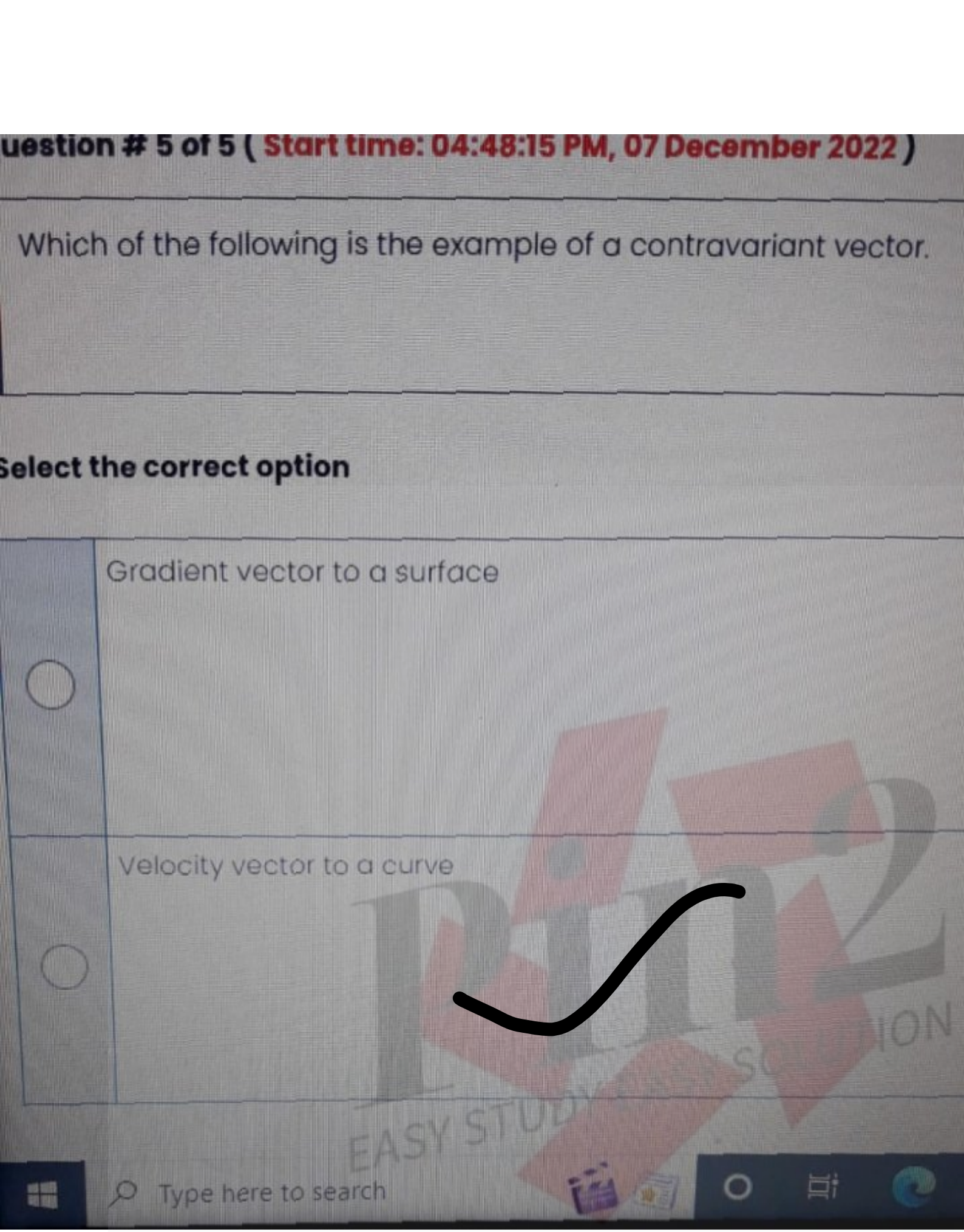
Type here to search

uestion # 5 of 5 (Start time: 04:48:15 PM, 07 December 2022)

Which of the following is the example of a contravariant vector.

Select the correct option

- Gradient vector to a surface
- Velocity vector to a curve



EASY STUDY

Type here to search

Question # 1 of 5 (Start time: 04:52:59 PM, 07 December 2022)

In Einstein notation, covariant components of a vector are denoted with_____

Select the correct option

- lower indices
- upper indices

PIT EASY STUDY EASY SOLUTION

Question # 3 *of* 5 (Start time: 04:55:12 PM, 07 December 2022)

In Einstein notation, covariant components of a vector are denoted with_____.

Select the correct option

- upper indices
- lower indices



EASY STUDY EASY SOLUTION

vulms.vu.edu.pk/

MTH623 - Tensor Analysis and its Applications (Quiz #1) Quiz Start Time: 04:

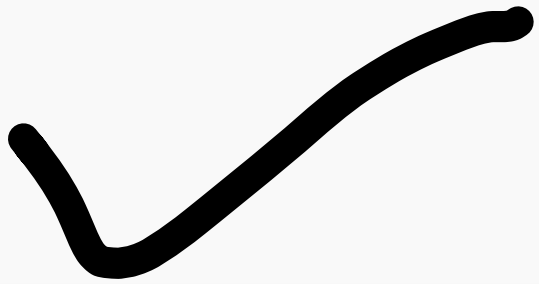
Question # 5 of 5 (Start time: 04:56:17 PM, 07 December 2022)

Covariant components of a vector represent its _____

contravariant in parallel

Select the correct option

- parallel projections
- perpendicular projections



PIU EASY STUDY EASY SOLUTION

4G 5:04 4.20 KB/s 64
vulms.vu.edu.pk/ 3

BC210426547: MUHAMMAD NAEEM

MTH623 - Tensor Analysis and its Applications (Quiz #1) Quiz Start

Question # 4 of 5 (Start time: 05:04:26 PM, 07 December 2022)

In Cartesian coordinate system the covariant and contravariant components of a vector are different.

Select the correct option

- true
- false

EASY STUDY EASY SOLUTION

Click to Save

Question # 4 of 5 (Start time: 05:16:49 PM, 07 December 2022)

In Einstein notation, covariant components of a vector are denoted with_____.

contravariant mn upper indices

Select the correct option

- ○ upper indices
- ◉ lower indices



EASY STUDY EASY SOLUTION

Question # 3 of 5 (Start time: 05:16:38 PM, 07 December 2022)

Using the transformation law $\tilde{x}_1 = 2x_1$ and $\tilde{x}_2 = x_2$ find $\frac{\partial x^1}{\partial \tilde{x}^1}$

Select the correct option

- (●) 1/2
- ( ) 1

PUZ EASY STUDY EASY SOLUTION

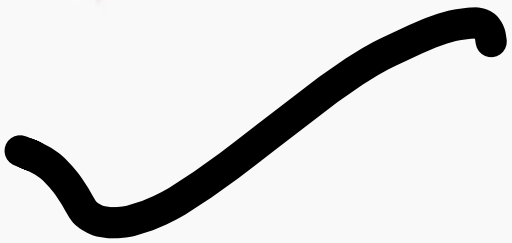
MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 2 of 5 (Start time: 05:16:29 PM, 07 December 2022)

If $\tilde{x}^1 = 2x^1$ and $\tilde{x}^2 = x^2$ then $x^1 =$_____ in terms of tilde coordinates.

Select the correct option

- ○ $2\tilde{x}^1$
- ● $\frac{1}{2}\tilde{x}^1$



EASY STUDY EASY SOLUTION

Which one of the following is the covaraint tensor of order 3?

Select the correct option

○	A_{kl}^{m}
●	T_{ijk}
○	T^{ijk} in contravariant this is true
○	A_{ij}^{j}

BC210414689: KIRAN MARIA

Time Left 87 sec(s)

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 05:16 PM, 07 December 2022

Question # 5 of 5 (Start time: 05:16:59 PM, 07 December 2022)

Total Marks: 1

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x^1}{\partial \tilde{x}^1}$?

Select the correct option

Reload Math Equations

- $\cos\theta$
- $-\cos\theta$

EASY STUDY EASY SOLUTION

estion # 3 of 5 (Start time: 05:01:06 PM, 07 December

The transformation rule for mixed tensor of order 2 is__

ect the correct option

$$\tilde{T}^{i}_{j} = \frac{\partial \tilde{x}^{i}}{\partial x^{m}} \frac{\partial x^{n}}{\partial \tilde{x}^{j}} T^{m}_{n}$$



$$\tilde{T}^{i}_{j} = \frac{\partial \tilde{x}^{m}}{\partial x^{i}} \frac{\partial x^{i}}{\partial \tilde{x}^{n}} T^{m}_{n}$$

uestion # 2 of 5 (Start time: 04:59:36 PM, 07 December 2022)

Using the transformation law $\tilde{x}_1 = 2x_1$ and $\tilde{x}_2 = x_2$ find $\frac{\partial \tilde{x}^1}{\partial x^2}$

elect the correct option

○	0
○	1
○	2
○	1/2

EASY STUDY EASY SOLUTION

Question # 5 of 5 (Start time: 04:56:17 PM, 07 December 2022)

Covariant components of a vector represent its _____

Select the correct option

- parallel projections
- perpendicular projections

EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 4 of 5 (Start time: 04:56:31 PM, 07 December 2022)

The transformation law for a covariant vector is given by

Select the correct option

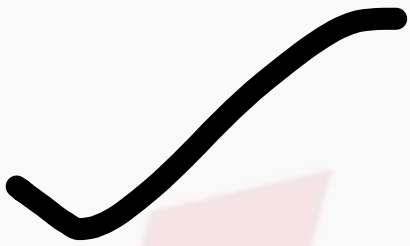
- $\bar{A}_i = A_j \frac{\partial x^j}{\partial \bar{x}^i}$
- $\bar{A}_i = A_i \frac{\partial x^j}{\partial \bar{x}^j}$

EASY STUDY EASY SOLUTION

Question # 1 of 5 (Start time: 04:58:02 PM, 07 December 2022)

A contravariant vector is often abbreviated as_____.

Select the correct option

- velocity vector 
- vector

EASY STUDY EASY SOLUTION

Question # 5 of 5 (Start time: 04:58:34 PM, 07 December 2022)

Total Marks: 1

On transforming a vector the_____ change and the_____change however, the vector itself does not change.

Select the correct option

Reload Math Equations

- components, unit vectors
- basis vectors, unit vectors
- magnitude, basis vectors
- components, basis vectors

Click to Save Answer & Move to Next Question

PU2 EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 4 of 5 (Start time: 04:57:13 PM, 07 December 2022)

_____ of a vector represent its parallel projections onto coordinate axes.

Select the correct option

- Contravariant components
- Covariant components

PUC EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 5 of 5 (Start time: 04:56:17 PM, 07 December 20

Covariant components of a vector represent its _____

Select the correct option

- parallel projections
- perpendicular projections

EASY STUDY EASY SOLUTION

Question # 5 of 5 (Start time: 08:07:25 PM, 07 December 2022) **Total Marks:** 1

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x^1}{\partial \tilde{x}^1}$?

Select the correct option

Reload Math Equations

○	$\cos\theta$
○	$-\cos\theta$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x'}{\partial r'}$?

Select the correct option

Reload Math Equations

- $-\cos\theta$
- $\cos\theta$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Question # 3 of 5 (Start time: 08:07:49 PM, 07 December 2022)

The transformation law for a covariant vector is given by

Select the correct option

○	$\bar{A}_i = A_i \frac{\partial x^i}{\partial \bar{x}^i}$
○	$\bar{A}_i = A_j \frac{\partial x^j}{\partial \bar{x}^i}$

MC220201510: RUBI NASEER

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Sta

Question # 5 of 5 (Start time: 08:08:54 PM, 07 December 2022)

Covariant components of a vector represent its _____

Select the correct option

- parallel projections
- perpendicular projections

Click to Save

MC220201510: RUBI NASEER

MTH623 - Tensor Analysis and Its Applications (Quiz #1)

Quiz Start Time: 08:0

Question # 4 of 5 (Start time: 08:08:18 PM, 07 December 2022)

In Einstein notation, covariant components of a vector are denoted with _____

Select the correct option

- lower indices
- upper indices

Click to Save Answer & N

Question # 2 of 5 (Start time: 08:07:38 PM, 07 December 2022)

_____ of a vector represent its parallel projections onto coordinate axes.

Select the correct option

- Covariant components
- Contravariant components

Which one of the following is the covariant tensor of order 3?

Select the correct option

Reload Math Equations

○	T_{ijk}
○	A_{ij}^{j}
○	A_{kl}^{m}
●	T^{ijk}

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

MC220201510: RUBI NASEER

Time Le

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 08:06 PM, 07 Dece

Question # 1 of 5 (Start time: 08:06:42 PM, 07 December 2022)

To

In Cartesian coordinate system the covariant and contravariant components of a vector are different

Select the correct option

Reload Math E

- false
- true

Click to Save Answer & Move to Next Que

EASY STUDY SOLUTION

Question # 4 of 5 (Start time: 08:06:36 PM, 07 December 2022) **Total Marks:** 1

The transformation law for a covariant vector is given by

Select the correct option

Reload Math Equations

○	$\tilde{A}_i = A_j \frac{\partial x^j}{\partial \tilde{x}^i}$
○	$\tilde{A}_i = A_i \frac{\partial x^j}{\partial \tilde{x}^j}$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Question # 3 of 5 (Start time: 08:06:04 PM, 07 December 2022) **Total Marks:** 1

Using the transformation law $\bar{x}_1 = 2x_1$ and $\bar{x}_2 = x_2$ find $\frac{\partial \bar{x}^1}{\partial x^2}$

Select the correct option

Reload Math Equations

○	1/2
○	1
○	2
○	0

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Question # 2 of 5 (Start time: 08:05:25 PM, 07 December 2022) **Total Marks:** 1

Which of the following is the example of a contravariant vector.

Select the correct option

Reload Math Equations

○	Velocity vector to a curve
○	Gradient vector to a surface

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Question # 1 of 5 (Start time: 08:04:13 PM, 07 December 2022) **Total Marks:** 1

A contravariant vector is often abbreviated as_____.

Select the correct option

Reload Math Equations

- vector
- velocity vector

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Using the transformation law $\bar{x}_1 = 2x_1$ and $\bar{x}_2 = x_2$ find $\frac{\partial x^1}{\partial \bar{x}^1}$

Select the correct option

Reload Math Equations

- 1/2
- 1

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION



Extension Detected!

Dear Student,

It is observed that you have installed following browser extension(s)!

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EASY STUDY

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MC210401061 TANZEELA ARSHAD

Time Left

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:50 PM, 07 Decemb

Question # 5 of 5 (Start time: 07:55:37 PM, 07 December 2022)

Total

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x^1}{\partial \bar{x}^1}$?

Select the correct option

Reload Math Equa

$\cos\theta$

$-\cos\theta$

Click to Save Answer & Move to Next Ques

EASY STUDY SOLUTION

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MC210401061: TANZEELA ARSHAD

Time Left 17 sec(s)

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:50 PM, 07 December 2022

Question # 4 of 5 (Start time: 07:53:52 PM, 07 December 2022)

Total Marks: 1

Which of the following is the example of a contravariant vector

Select the correct option

Reload Math Equations

- Gradient vector to a surface
- Velocity vector to a curve

Click to Save Answer & Move to Next Question

EASY STUDY ASSOCIATION

7:54

Quiz

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MC210401061: TANZEELA ARSHAD

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:50 PM, 0

Question # 4 of 5 (Start time: 07:53:52 PM, 07 December 2022)

Which of the following is the [illegible] of a contravariant vector

Select the correct option

Gradient vector to a surface

Reload qu

Velocity vector to a curve

Click to Save Answer & Move to Next Q

7:54

Quiz
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MC210401961: TANZEELA ARSHAD

Time Left 73 sec(s)

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:50 PM, 07 December 2022

Total Marks: 1

Question # 4 of 5 (Start time: 07:53:52 PM, 07 December 2022)

Which of the following is an example of a contravariant vector

Select the correct option

Reload Math Equations

Gradient vector to a surface

Velocity vector to a curve

Click to Save Answer & Move to Next Question

EASY STUDY SOLUTION

Quiz
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MC210401061: TANZEELA ARSHAD

Tim

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:50 PM, 07 De

Question # 3 of 5 (Start time: 07:53:06 PM, 07 December 2022)

The transformation law for a contravariant vector is given by

Select the correct option

Reload Math

- $\vec{A} = A^i \frac{\partial x^i}{\partial x^j}$
- $\vec{A}^i = A^i \frac{\partial x^i}{\partial x^j}$

Click to Save Answer & Move to Next Qu

Quiz
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MC210401061 TANZEELA ARSHAD

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:50 PM

Question # 2 of 5 (Start time: 07:52:03 PM, 07 December 2022)

Which one of the following is the contravariant tensor of order 2

Select the correct option

Reload

- A^{ij}
- T^{ij}
- A^{i}_{j}
- T_{ij}

Click to Save Answer & Move to Next

MC210401061: TANZEELA ARSHAD

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 2 of 5 (Start time: 07:52:03 PM, 07 December 2022)

Which one of the following is the contravariant tensor of order 2.

Select the correct option:

- A^{ij}
- T^{ij}
- A_{ij}
- T_{ij}

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MC210401061: TANZEELA ARSHAD

Time Left 82

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:50 PM, 07 December 2022

Question # 1 of 5 (Start time: 07:50:46 PM, 07 December 2022)

Total Marks: 1

Covariant components of a vector represent its .....

Select the correct option

Reload Math Equations

- perpendicular projections
- parallel projections

Click to Save Answer & Move to Next Question

EASY STUDY SOLUTION

BC210417713: SABER ALI

Time

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:43 PM, 07 De

Question # 5 of 5 (Start time: 07:46:37 PM, 07 December 2022)

Which of the following is the example of a contravariant vector.

Select the correct option

Reload Mat

- Gradient vector to a surface
- Velocity vector to a curve

EASY STUDY EASY SOLUTION

Click to Save Answer & Move to Ne

Search the web and Windows

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BC210417713: SABER ALI

Time Left 87 sec(s)

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:43 PM, 07 December 2022

Question # 4 of 5 (Start time: 07:46:11 PM, 07 December 2022)

Total Marks: 1

Which one of the following is the covaraint tensor of order 3?

Select the correct option

Reload Math Equations

- A_{kl}^{m}
- A_{ij}^{j}
- T_{ijk}
- T^{ijk}

Click to Save Answer & Move to Next Question

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:43 PM, 07 December 20

Total Mar

Question # 3 of 5 (Start time: 07:44:56 PM, 07 December 2022)

---------- of a vector represent its parallel projections onto coordinate axes.

Reload Math Equatio

Select the correct option

- Covariant components
- Contravariant components

Click to Save Answer & Move to Next Q

Search the web and Windows

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 07:43 PM, 07 December 2022

Question # 2 of 5 (Start time: 07:44:43 PM, 07 December 2022)

Total Marks: 1

Covariant components of a vector represent its _____

Select the correct option

Reload Math Equations

- parallel projections
- perpendicular projections

Click to Save Answer & Move to Next Question

EASY STUDY SOLUTION

SABER ALI

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Sta

Question # 1 of 5 (Start time: 07:43:45 PM, 07 December 2022)

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then

Select the correct option

- ○ $\cos\theta$
- ○ $-\cos\theta$

EASY STUDY SOLUTION

Click to Save

Search the web and Windows

A vector whose _____transform in a contravariant fashion is a contravariant vector.

Select the correct option

Reload Math Equations

○	direction
○	components

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Question # 3 of 5 (Start time: 07:38:39 PM, 07 December 2022)

On transforming a vector the_____ change and the_____

Select the correct option

○	magnitude, basis vectors
○	components, basis vectors
○	basis vectors, unit vectors
○	components, unit vectors

Which one of the following is the contravariant tensor of order 2.

Select the correct option

Reload Math Equations

○	T_{ij}
○	A_k^l
○	A^{ii}
○	T^{ij}

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

_____ of a vector represent its parallel projections onto coordinate axes.

Select the correct option

Reload Math Equations

- Covariant components
- Contravariant components

Click to Save Answer & Move to Next Question

Question # 2 of 5 (Start time: 07:37:32 PM, 07 December 2022)

Which one of the following is the contravariant tensor of order 2.

Select the correct option

○	A_k^l
○	T_{ij}
○	T^{ij}
○	A^{ii}

In Cartesian coordinate system the covariant and contravariant components of a vector are different.

Select the correct option

Reload Math Equations

- True
- false

Click to Save Answer & Move to Next Question

Which of the following is the example of a contravariant vector.

Select the correct option

Reload Math Equations

○	Velocity vector to a curve
○	Gradient vector to a surface

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Question # 4 of 5 (Start time: 07:33:02 PM, 07 December 2022)

In Einstein notation, covariant components of a vector are denoted with___

Select the correct option

○	upper indices
○	lower indices

Covectors are also called_____.

Select the correct option

○	dual vectors
○	unit vectors

Using the transformation law $\tilde{x}_1 = 2x_1$ and $\tilde{x}_2 = x_2$ find $\frac{\partial \tilde{x}^1}{\partial x^1}$

Select the correct option

○	1
○	2
○	0
○	1/2

EASY STUDY EASY SOLUTION

On transforming a vector the_____ change and the_____change however, the vector itself does not change.

Select the correct option

Reload Math Equ

○	basis vectors, unit vectors
○	magnitude, basis vectors
○	components, unit vectors
○	components, basis vectors

Click to Save Answer & Move to Next Que

EASY STUDY EASY SOLUTION

Question # 5 of 5 (Start time: 07:07:52 PM, 07 December 2022)

the transformation law for a contravariant vector is given by

Select the correct option

$\vec{A}' = A' \frac{\partial x^i}{\partial x^j}$

$\vec{A}' = A' \frac{\partial x^j}{\partial x^i}$

Question # 4 of 5 (Start time: 07:05:50 PM, 07 December 2022)

The transformation rule for mixed tensor of order 2 is_____

Select the correct option

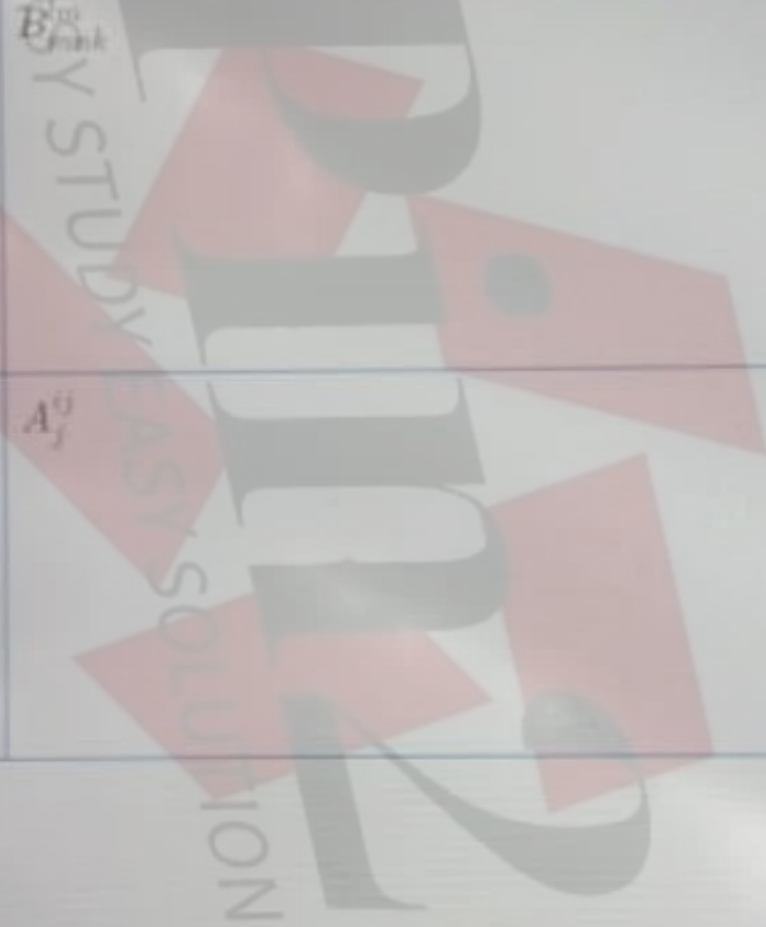
$\tilde{T}^i_j = \frac{\partial \tilde{x}^i}{\partial x^m} \frac{\partial x^n}{\partial \tilde{x}^j} T^m_n$

$\tilde{T}^i_j = \frac{\partial \tilde{x}^m}{\partial x^i} \frac{\partial x^n}{\partial \tilde{x}^j} T^m_n$

Question # 3 of 5 (Start time: 07:04:58 PM, 07 December 2022)

The order of _____ is 3

Select the correct option



Quiz
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Using the transformation law $\tilde{x}_1 = 2x_1$ and $\tilde{x}_2 = x_2$ find $\frac{\partial \tilde{x}^1}{\partial x^2}$

Select the correct option

	2
	1
	1/2
	0

EASY STUDY SOLUTION

Question # 1 of 5 (Start time: 07:03:09 PM, 07 December 2022)

_____ is a mixed tensor of order 3

Select the correct option

	T^{l}_{mn}
	T^{ij}_{j}
	T^{l}_{ln}
	T^{k}_{m}

EASY STUDY SOLUTION

Which of the following statement(s) is/are true?

1. A liquid does not form a free surface.
2. Gas expands to fill the entire available space.

Select the correct option

○	1 only
○	Neither 1 nor 2
○	1 and 2 both
○	2 only

Click to Save Answer & Mo

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordi

Select the correct option

○	$-r \sin \theta$
○	$r \sin \theta$

EASY STUDY EASY SOLUTION

Question # 2 of 5 (Start time: 07:25:51 PM, 07 December 2022)

Covariant components of a vector represent its _____

Select the correct option

○	perpendicular projections
○	parallel projections

_____ of a vector represent its parallel projections onto coordinate axes.

Select the correct option

○	Covariant components
○	Contravariant components

Question # 5 of 5 (Start time: 06:39:34 PM, 07 December 2022) Total Marks: 1

The transformation law for a contravariant vector is given by

Select the correct option

Reload Math Equations

- $\bar{A}^i = A^i \frac{\partial \bar{x}^j}{\partial x^j}$
- $\bar{A}^i = A^j \frac{\partial \bar{x}^i}{\partial x^j}$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Which one of the following is the covaraint tensor of order 3?

Select the correct option

Reload Math Equations

○	T_{ijk}
○	A_{kl}^{m}
○	A_{ij}^{j}
○	T^{ijk}

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

vulms.vu.edu.pk/

BC210407871: KIRAN SHAHZADI

Time Left 56 sec(s)

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 06:32 PM, 07 December 2022

Question # 4 of 5 (Start time: 06:34:42 PM, 07 December 2022)

Total Marks: 1

The transformation rule for mixed tensor of order 2 is_____.

Select the correct option

Reload Math Equations

EASY STUDY EASY SOLUTION

- $\bar{T}^{i}_{j} = \frac{\partial \bar{x}^{i}}{\partial x^{m}} \frac{\partial x^{n}}{\partial \bar{x}^{j}} T^{m}_{n}$
- $\bar{T}^{i}_{j} = \frac{\partial \bar{x}^{m}}{\partial x^{i}} \frac{\partial x^{i}}{\partial \bar{x}^{n}} T^{m}_{n}$

The transformation rule for mixed tensor of order 2 is_____.

Select the correct option

Reload Math Equations

- $\bar{T}^i_j = \frac{\partial \bar{x}^i}{\partial x^m} \frac{\partial x^n}{\partial \bar{x}^j} T^m_n$
- $\bar{T}^i_j = \frac{\partial \bar{x}^m}{\partial x^i} \frac{\partial x^i}{\partial \bar{x}^n} T^m_n$

Click to Save Answer & Move to Next Question

Using the transformation law $\tilde{x}_1 = 2x_1$ and $\tilde{x}_2 = x_2$ find $\frac{\partial x^1}{\partial \tilde{x}^1}$

Select the correct option

Reload Math Equations

○	1
○	1/2

Click to Save Answer & Move to Next Question

If tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x^1}{\partial \tilde{x}^1}$?

Select the correct option

Reload Math Equations

○	$\cos\theta$
○	$-\cos\theta$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Select the correct option

Reload Math Equations

○	Velocity vector to a curve
○	Gradient vector to a surface

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

A vector whose _____transform in a contravariant fashion is a contravariant vector.

Select the correct option

Reload Math Equations

○	components
○	direction

Click to Save Answer & Move to Next Question

Which one of the following is the covaraint tensor of order 3?

Select the correct option

Reload Math Equations

○	A_{kl}^{m}
○	T_{ijk}
○	T^{ijk}
○	A_{ij}^{j}

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Select the correct option

Reload Math Equations

○	A^{ij}
○	A_k^l
○	T_{ij}
○	T^{ij}

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

The transformation law for a contravariant vector is given by

ect the correct option

Reload Math Equation

- $\tilde{A}^i = A^j \frac{\partial x^i}{\partial x^j}$
- $\tilde{A}^i = A^i \frac{\partial x^j}{\partial x^i}$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

In Cartesian coordinate system the covariant and contravariant components of a vector are different.

Select the correct option

Reload Math Equations

- true
- false

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Reload Math Equations

○	upper indices
○	lower indices

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

_____ of a vector represent its parallel projections onto coordinate axes.

Select the correct option

- Covariant components
- Contravariant components

Select the correct option

Reload Math Equations

○	$\cos\theta$
○	$-\cos\theta$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Question # 5 of 5 (Start time: 06:25:18 PM, 07 December 2022)

If $\tilde{x}^1 = 2x^1$ and $\tilde{x}^2 = x^2$ then $x^1 =$_____ in terms of tilde coordinates.

Select the correct option

○	$\frac{1}{2}\tilde{x}^1$
○	$2\tilde{x}^1$

EASY STUDY EASY SOLUTION

Select the correct option

○	A^m_{kl}
○	A^j_{ij}
○	T_{ijk}
○	T^{ijk}

Click to S

_____ is a mixed tensor of order 3.

Select the correct option

Reload Math Equations

○	T_m^k
○	T_j^{ij}
○	T_{mn}^l
○	T_{ln}^i

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Select the correct option

○	true
○	false

Click to Save A

stion # 2 of 5 (Start time: 06:23:20 PM, 07 December 2022)

tilde coordinate system corresponds to polar coordinates and non tilde Cartesian coordinates then find $\frac{\partial x^1}{\partial \tilde{x}^1}$?

ct the correct option

- $-\cos\theta$
- $\cos\theta$

Click to Save Answ

Covectors are also called_____

Select the correct option

Reload Math Equations

○	dual vectors
○	unit vectors

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

Which statement(s) is(are) true about the following sequence $a_0 = 1$ and $a_n = 1 - e^{-a_{n-1}}$

Select the correct option

Reload Math Equations

- $0 < a_n \leq 1$ for all n.
- Sequence is convergent
- All of the above
- $a_{n+1} - a_n = -\left(e^{-a_n} - e^{-a_{n-1}}\right)$

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

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BC210407344: MADEEHA RUBAB

Time Left 86 sec(s)

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 08:16 PM, 07 December 2022

Question # 5 of 5 (Start time: 08:20:23 PM, 07 December 2022)

Total Marks: 1

_____ of a vector represent its parallel projections onto coordinate axes.

Select the correct option

Reload Math Equations

- Covariant components
- Contravariant components

PU2 EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 1 of 5 (Start time: 08:20:34 PM, 07 December 2022)

A vector whose _____transform in a contravariant fashion is a contravariant vector.

Select the correct option

○	components
○	direction

EASY STUDY EASY SOLUTION

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MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 1 of 5 (Start time: 08:20:34 PM, 07 December 2022)

A vector whose _____transform in a contravariant fashion is a contravariant vector.

Select the correct option

○	components
○	direction

EASY STUDY EASY SOLUTION

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Question # 2 of 5 (Start time: 08:21:28 PM, 07 December 2022)

Covariant components of a vector represent its _____

Select the correct option

○	perpendicular projections
○	parallel projections

VIRTUAL STUDY SOLUTION

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BC210427623: NOUREEN JAVAD

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 3 of 5 (Start time: 08:22:22 PM, 07 December 2022)

A contravariant vector is often abbreviated as_____.

Select the correct option

○	vector
○	velocity vector

EASY STUDY ONLINE

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BC210427623: NOUREEN JAVAD

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 3 of 5 (Start time: 08:22:22 PM, 07 December 2022)

A contravariant vector is often abbreviated as_____.

Select the correct option

○	vector
○	velocity vector

EASY STUDY ONLINE

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BC210427623: NOUREEN JAVAD

MTH623 – Tensor Analysis and its Applications (Quiz #1)

Question # 4 of 5 (Start time: 08:23:33 PM, 07 December 2022)

Using the transformation law $\tilde{x}_1 = 2x_1$ and $\tilde{x}_2 = x_2$ find $\frac{\partial \tilde{x}^1}{\partial x^2}$

Select the correct option

○	2
○	0
○	1
○	1/2

EASY STUDY ONLINE

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_____ is a mixed tensor of order 3.

Select the correct option

Reload Math Equations

○	T_{lm}^{i}
○	T_{m}^{k}
●	T_{j}^{ij}
○	T_{mn}^{i}

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION

BC210427623: NOUREEN JAVAD

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 3 of 5 (Start time: 08:22:22 PM, 07 December 2022)

A contravariant vector is often abbreviated as_____.

Select the correct option

○	vector
○	velocity vector

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BC210407344: MADEEHA RUBAB

Time Left 86 sec(s)

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Quiz Start Time: 08:16 PM, 07 December 2022

Question # 5 of 5 (Start time: 08:20:23 PM, 07 December 2022)

Total Marks: 1

_____ of a vector represent its parallel projections onto coordinate axes.

Select the correct option

Reload Math Equations

- Covariant components
- Contravariant components

PU2 EASY STUDY EASY SOLUTION

MTH623 - Tensor Analysis and its Applications (Quiz #1)

Question # 1 of 5 (Start time: 08:20:34 PM, 07 December 2022)

A vector whose _____transform in a contravariant fashion is a contravariant vector.

Select the correct option

○	components
○	direction

EASY STUDY EASY SOLUTION

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Question # 2 of 5 (Start time: 08:21:28 PM, 07 December 2022)

Covariant components of a vector represent its _____

Select the correct option

○	perpendicular projections
○	parallel projections

VIRTUAL STUDY SOLUTION

Type here to search

Which of the following is the example of a contravariant vector.

Select the correct option

Reload Math Equations

○	Gradient vector to a surface
○	Velocity vector to a curve

Click to Save Answer & Move to Next Question

EASY STUDY EASY SOLUTION